

## LOME, Togo

WMO: 653870

Lat: 6.166N

Lon: 1.255E

Elev: 72

StdP: 14.66

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 71.2       | 72.3 | 50.2                          | 53.9 | 83.6 | 58.9 | 74.4 | 80.7 | 20.7                  | 81.7 | 19.4 | 81.5 | 3.9                   | 0    | 0.397 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 3             | 11.3                   | 91.8            | 78.9 | 91.4 | 79.4 | 90.6 | 79.5 | 82.8                | 87.4 | 82.3 | 86.9 | 81.8 | 86.5 | 10.5                 | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 82.1                            | 167.8 | 84.8 | 81.0 | 161.3 | 84.5 | 80.7 | 160.0 | 84.5 | 46.9          | 88.0 | 46.4 | 86.5 | 45.7 | 87.0 | 86.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 18.6              | 17.1 | 15.9 | DB                         | 67.8 | 96.0               | 1.8 | 2.3                                                | 66.5 | 97.7       | 65.4 | 99.0       | 64.4 | 100.3      | 63.1 | 102.0 |
|                   |      |      | WB                         | 58.3 | 84.8               | 4.0 | 1.3                                                | 55.4 | 85.8       | 53.0 | 86.5       | 50.7 | 87.2       | 47.8 | 88.2  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 82.2        | 82.8  | 84.6  | 84.9  | 84.3  | 82.9  | 80.5  | 79.3  | 78.7  | 80.0  | 81.5  | 83.3  | 83.3  |
|                                                                        | DBStd       | 2.72        | 2.17  | 1.83  | 1.99  | 2.25  | 2.26  | 1.93  | 1.36  | 1.29  | 1.47  | 1.61  | 1.43  | 1.71  |
|                                                                        | HDD50       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD50       | 11735       | 1016  | 968   | 1082  | 1028  | 1021  | 914   | 908   | 890   | 901   | 976   | 998   | 1033  |
|                                                                        | CDD65       | 6260        | 551   | 548   | 617   | 578   | 556   | 464   | 443   | 425   | 451   | 511   | 548   | 568   |
|                                                                        | CDH74       | 65766       | 6026  | 6618  | 7684  | 7048  | 6405  | 4356  | 3374  | 2765  | 3698  | 5042  | 6279  | 6471  |
|                                                                        | CDH80       | 22936       | 2276  | 2812  | 3413  | 3021  | 2447  | 1042  | 496   | 354   | 702   | 1431  | 2376  | 2566  |
| (14) Wind                                                              | WSAvg       | 7.7         | 6.3   | 8.1   | 8.5   | 7.8   | 6.4   | 7.1   | 9.4   | 10.4  | 9.1   | 7.1   | 6.3   | 5.9   |
| (15) Precipitation                                                     | PrecAvg     | 33.7        | 0.3   | 0.8   | 2.1   | 4.0   | 5.9   | 8.7   | 3.5   | 1.2   | 2.6   | 3.3   | 0.9   | 0.3   |
|                                                                        | PrecMax     | 60.0        | 2.4   | 6.7   | 6.1   | 8.3   | 15.5  | 27.6  | 12.5  | 6.4   | 16.0  | 9.1   | 4.5   | 1.7   |
|                                                                        | PrecMin     | 17.3        | 0.0   | 0.0   | 0.0   | 0.0   | 1.0   | 1.7   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 10.2        | 0.6   | 1.4   | 1.5   | 2.0   | 3.1   | 5.4   | 3.2   | 1.5   | 2.8   | 2.4   | 0.9   | 0.5   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 93.1  | 93.3  | 93.2  | 92.9  | 91.7  | 89.6  | 86.2  | 85.8  | 87.8  | 89.8  | 91.8  | 92.2  |
|                                                                        |             | MCWB        | 69.8  | 75.4  | 80.1  | 81.3  | 81.0  | 80.4  | 78.3  | 77.2  | 79.0  | 79.8  | 80.4  | 75.7  |
|                                                                        | 2%          | DB          | 91.0  | 91.6  | 91.7  | 91.5  | 90.1  | 87.7  | 84.6  | 84.3  | 86.2  | 88.2  | 90.1  | 91.2  |
|                                                                        |             | MCWB        | 75.1  | 78.7  | 80.6  | 81.1  | 80.5  | 79.9  | 77.6  | 76.8  | 78.5  | 79.4  | 80.0  | 77.3  |
|                                                                        | 5%          | DB          | 89.7  | 90.7  | 91.1  | 90.0  | 89.5  | 86.2  | 84.0  | 82.8  | 84.7  | 87.6  | 89.6  | 89.9  |
|                                                                        |             | MCWB        | 76.8  | 79.2  | 80.6  | 80.7  | 80.4  | 79.1  | 77.3  | 76.1  | 77.7  | 79.2  | 79.9  | 78.2  |
|                                                                        | 10%         | DB          | 88.2  | 89.6  | 89.7  | 89.5  | 88.1  | 84.6  | 82.6  | 82.2  | 84.0  | 86.1  | 88.2  | 89.3  |
|                                                                        |             | MCWB        | 77.5  | 79.3  | 80.2  | 80.5  | 80.2  | 78.6  | 76.6  | 75.8  | 77.4  | 78.6  | 79.7  | 78.4  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 82.4  | 83.2  | 83.2  | 83.7  | 82.9  | 82.2  | 80.0  | 79.0  | 80.5  | 81.5  | 82.6  | 82.6  |
|                                                                        |             | MCDB        | 86.3  | 87.8  | 88.3  | 89.1  | 87.8  | 86.0  | 83.2  | 82.7  | 85.3  | 86.9  | 87.8  | 87.1  |
|                                                                        | 2%          | WB          | 81.3  | 82.4  | 82.5  | 82.7  | 82.1  | 80.8  | 79.1  | 77.9  | 79.2  | 80.3  | 81.6  | 81.6  |
|                                                                        |             | MCDB        | 85.4  | 86.6  | 87.0  | 87.5  | 87.1  | 84.9  | 82.4  | 81.7  | 83.7  | 85.6  | 87.0  | 86.3  |
|                                                                        | 5%          | WB          | 80.7  | 81.8  | 81.9  | 82.0  | 81.4  | 80.0  | 78.4  | 77.2  | 78.7  | 79.7  | 80.9  | 80.8  |
|                                                                        |             | MCDB        | 84.9  | 86.1  | 86.5  | 86.7  | 86.4  | 84.0  | 81.7  | 81.1  | 83.0  | 84.7  | 86.4  | 85.8  |
|                                                                        | 10%         | WB          | 80.1  | 81.1  | 81.5  | 81.5  | 80.7  | 79.2  | 77.7  | 76.6  | 78.1  | 79.1  | 80.3  | 80.2  |
|                                                                        |             | MCDB        | 84.3  | 85.7  | 86.3  | 86.4  | 85.8  | 83.1  | 81.0  | 80.2  | 82.2  | 83.8  | 85.6  | 85.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 14.0  | 12.0  | 11.3  | 11.4  | 11.3  | 9.5   | 8.2   | 8.3   | 9.2   | 11.1  | 12.5  | 13.9  |
|                                                                        |             | MCDBR       | 14.8  | 12.7  | 11.7  | 11.8  | 12.1  | 11.1  | 9.4   | 9.1   | 10.1  | 12.2  | 13.1  | 14.3  |
|                                                                        | 5% WB       | MCWBR       | 6.6   | 5.4   | 4.8   | 5.1   | 5.2   | 5.5   | 4.3   | 4.3   | 4.7   | 5.3   | 5.4   | 5.9   |
|                                                                        |             | MCWBR       | 11.8  | 11.1  | 11.1  | 11.3  | 11.5  | 10.3  | 8.5   | 8.5   | 9.9   | 11.6  | 12.7  | 12.7  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.817 | 0.817 | 0.812 | 0.646 | 0.544 | 0.519 | 0.527 | 0.525 | 0.488 | 0.501 | 0.589 | 0.698 |
|                                                                        |             | taud        | 1.520 | 1.530 | 1.533 | 1.838 | 2.072 | 2.159 | 2.128 | 2.131 | 2.262 | 2.218 | 1.932 | 1.709 |
|                                                                        | Ebn at Noon |             | 184   | 190   | 193   | 225   | 243   | 246   | 245   | 250   | 264   | 261   | 235   | 207   |
|                                                                        |             | Edh at Noon | 93    | 94    | 94    | 68    | 53    | 47    | 49    | 50    | 45    | 47    | 61    | 75    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1619  | 1616  | 1691  | 1702  | 1566  | 1343  | 1410  | 1419  | 1514  | 1660  | 1681  | 1613  |
|                                                                        | RadStd      |             | 50    | 69    | 77    | 69    | 85    | 81    | 93    | 89    | 62    | 75    | 44    | 46    |

## Historical Trends

|                             | DBAvg | Heating |       |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |       | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)   | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |       | N/A     | N/A   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (4 neighbors) |       | +0.50   | +0.56 | N/A    | +0.65   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page